

Soumyajyoti Dutta

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Empirical LLM Research | Post-Training & Evaluation | AI for Cyberdefense & Adversarial ML

EXPERTISE

Core Competencies:

- **LLM post-training:** supervised fine-tuning and curriculum-staged training to align foundation models (135M–~9B, incl. sparse MoE) to intended behavior, e.g. teaching an agent to clarify rather than hallucinate on ambiguous intent.
- **Evaluation & diagnosis:** task-specific metrics (lexical + syntactic + SMT-based semantic comparison), LLM-as-judge and user-simulation harnesses as training-time evaluation signals; frontier-API baselines and stage-wise ablations that trace *where* a method's gains stop and *why*.
- **AI × security:** LLM systems for automated cyberdefense (threat-intel rule generation, IOC/TTP extraction), adversarial ML (built a competition's top-ranked adversarial attack suite), and ML malware detection on 2M+ samples.
- **Efficient training & inference:** single-GPU and multi-node SLURM training with FFT and PEFT (LoRA/QLoRA), MoE, grammar-derived hybrid tokenization, and vLLM inference with multi-turn KV reuse.

Technical Stack:

Python (PyTorch, HuggingFace Transformers, TRL, TensorFlow, NumPy, Pandas, scikit-learn, Matplotlib), SQL/SQLite, DeepSpeed, FSDP, DDP/NCCL, CUDA, SLURM, vLLM, Z3/SMT, MCP, Ollama, Playwright, Docker, AWS, Git, CI/CD, FastAPI/Flask/Django, React, C++, Rust

PROFESSIONAL EXPERIENCE

Texas A&M University — Botacin's Lab

College Station, TX

Research Assistant / Machine Learning Engineer (Applied AI)

Spring 2024 – Present

- Built end-to-end LLM post-training pipelines (data curation, SFT and curriculum training, evaluation, deployment) across a dozen model families (135M–~9B) on multi-node SLURM/GPU clusters.
- **YaraGen**, sponsored by **Laboratory for Physical Sciences, UMD & Booz Allen Hamilton**: grammar-oriented minimal LLMs for YARA rule generation (**submitted, USENIX Security '27, Cycle 1**):
 - Trained a **139M-parameter BART** generator to **91.8% semantic correctness** (93.7% in-window) on 22.7K held-out real-world rules vs. 93.8% for a fully fine-tuned 8B: **57× fewer parameters**, 17–20× the throughput.
 - Benchmarked frontier APIs: fully fine-tuned Llama and Ministral 8B models beat few-shot GPT-5.4 (84.0%) by **4–8 pp**, and zero-shot GPT-5.4 (73.5%) and Sonnet 5 (75.0%) by more; the 139M pipeline leads few-shot GPT-5.4 by ~10 pp. Frontier parse rates of 97–99.8% put the gap in semantics, not syntax.
 - Ran **150+ post-training runs** (FFT, LoRA/QLoRA) over seq2seq models (BART, CodeT5p, T5Gemma) and causal models (SmolLM2; LFM2.5, 230M–1.2B; Gemma 3, 270M–1B; Llama 3.1 8B / 3.2 1B; Ministral 3 8B), including seq2seq-vs-causal ablations.
 - Designed the **YaraAST hybrid tokenizer**: grammar-derived atomic tokens for rule structure over base subwords for IOC content: +4 pp accuracy, 11% shorter per-rule context, 5% faster training (3.8% fewer FLOPs, 16 GPUs).
 - Co-designed **Scooper**, a 55M BiLSTM NER model that masks IOCs behind placeholders (99.4% precision / 99.6% recall), adding up to +10 pp correctness and decoupling IOC size from the context window.
 - Trained **LLMScooper**, a LoRA alternative to Scooper (r=32, ~85M params) that turns the 8B generator's own backbone into an IOC tagger at **0.999 span F1**, swapped on GPU-resident weights.
 - Assembled three corpora (**305K crawled** in-the-wild rules from 59 repos; **15M synthetic** from 459 curated templates; **15M grammar-fuzzed**) and built the *Brownie & Puff* composite evaluators (BLEU + parse/compile + **Z3 SMT semantic comparison**) as training-time signals; only the composite correlated positively with test accuracy (BLEU Spearman down to -0.70).
 - Downstream on EMBER malware clusters, generated rules reach **96±4% TP at 0.8% FP** (99.7% generation success) vs. AutoYara's 93±6% at 0.5% FP; open-sourced the framework, models, and an agentic web chatbot.
 - *In progress*: extending YaraGen with tool use and verify-and-repair; designing verified multi-task SFT and an RLVR pilot with executable, formally checked rewards.
- **IOCScooper++**: IOC/TTP extraction from threat reports (**targeting USENIX Security '27, Cycle 2**):
 - Trained **ScooperNER**, a 50.6M-parameter extractor (MLM pretraining → fine-tuning, 36 IOC types): **0.894 micro / 0.925 macro** span-exact F1 on held-out vendor and government reports, beating fine-tuned encoder baselines 3–7× its size, with 0.950 F1 on explicit MITRE ATT&CK IDs.
 - End to end on 47 held-out reports, extraction feeds frozen YaraGen at 0.86/0.88 precision/recall with **100% rule validity** at **0.97 IOC coverage** across 344 generated rules.
 - Extending to grounded (entity, action, IOC) tuples mapped open-set onto all 637 ATT&CK techniques via procedure-example retrieval, with a co-annotated benchmark of real malware reports.

- **AutoPYara**: shipped an open-source rule-generation package (**PyPI**; **ACSAC 2026, accepted**) for $\approx 71\text{K}$ Windows PE samples: **+14 pp** coverage, 7.1 pp lower variance, and +10% threat-hunting accuracy over AutoYara (prior SOTA).

Cognizant Technology Solutions

Kolkata, India

*Junior Software Engineer — Client-Embedded Consulting (Client: **TJX Companies**)*

Aug 2021 – Aug 2022

- Client-embedded consultant for **TJX Companies (Fortune 500)**: shipped a logistics-automation platform; optimized Python/REST services (latency, concurrency, CI/CD) and led cross-functional code reviews.

Software Engineering Intern — converted to full-time offer

Feb 2021 – Jun 2021

- Built web features for a client logistics platform; resolved support tickets via ServiceNow triage and Splunk tracing.

SELECTED PROJECTS

Ambiguity-Aware Research Agent | *Co-authored (manuscript in prep.); led model development & training* 2026

- Aligned a deep-research agent to ask a clarifying question on underspecified intent instead of answering the wrong question, via four-stage SFT of Gemma-4-E4B (sparse MoE, $\sim 9\text{B}/4\text{B}$) on one 80GB GPU with vLLM.
- Built a three-axis evaluation with GPT-4o user simulation and LLM-as-judge; lifted pause-detection F1 from **0.168 to 0.573** and showed clarified answers still trail the no-pause baseline, isolating where SFT alone falls short.

Depixelation Recoverability Frontier | *Personal vision-security research, in progress* 2026

- Designing an open-source benchmark of when pixelated text is recoverable: a **360-cell** difficulty grid with a **10-seed** protocol that recasts depixelation as constrained search with an exact render-and-verify check, plus retrieval-augmented recognition with calibrated abstention on $4\times\text{H}100$.

Sturdy Fishstick: Full-Stack Applied-LLM Platform | *FastAPI, React, SQLite, local LLMs via Ollama* 2025

- Deployed a self-hosted job and PhD search platform that has scored **~ 750 listings** from 10+ sources, using categorical scoring that makes 1–2B local models reliable judges and FTS5-grounded RAG that rules out hallucinated counts.
- Built ingestion over 6 ATS JSON APIs, LinkedIn, and Google Jobs, with a Playwright fallback.

Multimodal & Vision Evaluation | *ECEN 642, Image Processing & Computer Vision* 2024

- Evaluated a DeepSeek vision-language model against human-annotated ground truth for object recognition and scene understanding; built the comparison harness and error analysis.

ML Malware Detection & Adversarial Attack Suite | *1st Place; still #1 after five offerings* 2023

- Built an end-to-end Windows PE malware classifier (EMBER/BODMAS/Benign-NET, 2M+ samples; LightGBM/XGBoost) at **96.2% accuracy, 0.97 F1**, served from a Docker HTTP endpoint; designed the best-performing adversarial attack suite in the CSCE 689 competition.

EDUCATION

Ph.D. in Computer Science, Texas A&M University Spring 2024 – Expected Spring 2028

- *Research focus*: LLM post-training, evaluation, and interpretability for automated cyberdefense.

M.S. in Computer Engineering, Texas A&M University Fall 2022 – Fall 2023

- *Coursework*: NLP, Deep Learning, Computer Vision, ML Cyberdefenses, ML Theory, Data Mining.

B.Tech. in Electronics & Communication Engineering, SRM Institute of Science and Technology 2017 – 2021

PUBLICATIONS & MANUSCRIPTS

YaraGen: minimal grammar-oriented LLMs for YARA rule generation USENIX Security '27, submitted (Cycle 1)

IOCScooper++: grounded IOC/TTP extraction from threat reports USENIX Security '27, in prep. (Cycle 2)

AutoPYara: automated YARA rule generation, open-sourced on PyPI ACSAC 2026, accepted

Proactive Deep Research Agents: ambiguity-aware clarification (co-authored) 2026, in preparation

Carotid-Artery Ultrasound Classification: CNN-based medical imaging Springer book chapter, 2022

TEACHING, TALKS & OUTREACH

Teaching Assistant, Texas A&M University: CSCE 313 and CSCE 439/704 2026

- CSCE 313 Computer Systems (Fall 2026): lead labs and write Linux assignments on processes, IPC, and threads.
- CSCE 439/704 Data Analytics for Cybersecurity (Spring 2026): taught clustering, supervised ML, and anomaly detection to 37 students; mentored 9 team projects.

Presenter, Texas Security Workshop: IOCScooper++, IOC and TTP extraction from threat reports 2026

Partner & Presenter, TAMU CS Day: interactive AI/cybersecurity demos for high-school students 2024, 2025